

Torin Chacko

Computational Mathematics & Systems Engineering

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Education

University of California, Los Angeles

B.S., Mathematics of Computation

Expected 2028

Los Angeles, CA

- Relevant coursework: CS33 Computer Architecture, Operating Systems, CS32 Data Structures & Algorithms, Linear Algebra, Real Analysis, Complex Analysis, Statistical Models
- Activities: ASME, ACM, Bruinrunners, IEEE, IDEAhacks

Experience

Software Engineer

Numberzoo

Jul 2026 – Present

Hybrid

- Connected client data with custom-built chat applications for small-business clients
- Built AI-powered workflow automation and automated document creation pipelines
- *Technologies: SQL, PostgreSQL, C#, .NET, Node.js, AI automation*

IST Systems Engineer Intern

County of Marin

Jun 2025 – Aug 2025

San Rafael, CA

- Updated out-of-date County systems in C#
- Worked with the LAND division to modernize their platform and tooling, expanding user access
- Learned and applied SQL, data systems, query design, and the .NET Framework
- *Technologies: C#, SQL, .NET Framework, HTML, CSS*

Projects

x86-64 Emulator

C++20, CMake

- Building a from-scratch x86-64 emulator and debugger covering instruction semantics, machine-code decoding, ELF loading, virtual memory, and syscalls
- Implemented CPU state and flag-correctness (CF/PF/AF/ZF/SF/OF) verified against native hardware with a dependency-free unit-test suite

Compartmental Human Metabolic Model

Python

- Built an hour-by-hour, ten-subsystem simulation of human metabolism responding to diet and exercise inputs
- Enforced strict double-entry mass and energy conservation across every organ compartment and environmental sink

Garmin Field Log

Python, GitHub Actions, HTML

- Connected the Garmin API to a serverless fitness dashboard: a scheduled GitHub Actions job pulls Garmin Connect data daily and a static GitHub Pages frontend renders it
- Created a backfill feature to sequentially pull all historical Garmin data and update the site

Skills

Languages

C, C++, C#, Python, x86-64 Assembly, MIPS, SQL

Tools & Frameworks

.NET Framework, Node.js, Git, CMake, Linux, MySQL, PostgreSQL, GitHub Actions, SolidWorks, ROS

Domains

Low-Level Systems, Cybersecurity, Algorithms